Singapore School Kinderland

Secondary 1 (Sec 1) Computer Science — Term 2 Comprehensive Practice Worksheet — TEACHER ANSWER KEY

Topics: Error Detection Methods (TB Pg 54–62) & Encryption (TB Pg 63–74) | Total Marks: 40 | Detailed Marking Scheme & Solutions

Student Name: _____ **Class:** Sec 1 | **Date:** _____ **Marks:** / 40

SECTION A: PARITY CHECKS & PARITY BLOCKS (10 MARKS)

Q1. Parity Bit Calculation *[4 Marks]*

A computer system transmits 7-bit ASCII data. Add the required Parity Bit (most significant bit) for each byte below:

Byte	7-bit Data Payload	Parity System	Required Parity Bit (0 or 1)	Complete 8-bit Transmitted Byte
A	1 1 0 0 1 0 1	Even Parity	**0 (count=4)**	**0 1 1 0 0 1 0 1**
B	1 0 1 1 1 1 0	Even Parity	**1 (count=5→6)**	**1 1 0 1 1 1 1 0**
C	0 1 1 0 0 0 1	Odd Parity	**0 (count=3)**	**0 0 1 1 0 0 0 1**
D	1 1 1 0 1 1 1	Odd Parity	**1 (count=6→7)**	**1 1 1 1 0 1 1 1**

Q2. Parity Block Matrix Analysis & Auto-Correction *[6 Marks]*

A block of four bytes was transmitted using **Even Parity**. An extra parity byte was appended to the bottom:

Row	Bit 7 (Parity)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Row Status (OK / Error)
Byte 1	1	0	1	1	0	0	1	0	**OK (4 ones)**
Byte 2	0	1	1	0	1	0	1	1	**ERROR (5 ones)**
Byte 3	1	1	0	1	1	0	0	1	**OK (4 ones)**
Byte 4	1	0	0	1	0	1	1	0	**OK (4 ones)**
Parity Byte	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**OK**

(a) Identify the corrupted row and column: **Byte 2** | Corrupted Column: **Bit 4** (Column 4 has 3 ones → odd) *[2]*

(b) Original bit: **0** | Corrected bit: **1** *[2]*

(c) Explain why this parity block cannot correct an error if two bits flip in the same row and column: *[2]*

Model Answer: If 2 bits flip in the same row and same column, both row parity and column parity will still register as even numbers, meaning the parity checks will report NO error and the corrupted bits cannot be identified.

SECTION B: CHECKSUMS, ECHO CHECKS & CHECK DIGITS (10 MARKS)

Q3. Checksum Verification Process *[3 Marks]*

Describe how a checksum algorithm ensures that a transmitted data file has arrived without corruption.

Q4. Echo Check Critique *[3 Marks]*

A student proposes using an Echo Check for online multiplayer gaming. Give TWO distinct reasons why an Echo Check is unsuitable for high-speed network applications.

Reason 1:

Reason 2:

Q5. Check Digits vs Transmission Checks *[4 Marks]*

(a) State TWO human data entry errors that check digits detect but parity checks do not. *[2]*

(b) The following 12 digits of an ISBN-13 barcode are scanned: `9 7 8 1 5 1 0 4 8 4 4 2`.
Calculate the final 13th Check Digit using the weighting `(1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3)`. Show your calculation: *[2]*

SECTION C: ARQ PROTOCOL SCENARIOS & TIMEOUTS (10 MARKS)

Q6. ARQ Protocol Operation *[5 Marks]*

The diagram below outlines an Automatic Repeat Request (ARQ) transmission sequence between Computer A and Computer B:

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Step 1: Computer A sends Packet 1 → starts Timeout Timer.
Step 2: Computer B receives Packet 1 → calculates Checksum → Checksum matches.
Step 3: Computer B sends [ Message X ] → Computer A cancels timer → sends Packet 2.
Step 4: Packet 2 gets corrupted by electrical noise in transit.
Step 5: Computer B detects checksum mismatch → sends [ Message Y ].
Step 6: Computer A receives Message Y → immediately re-transmits [ Packet Z ].
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(a) Identify Message X: _____ *[1]*

(b) Identify Message Y: _____ *[1]*

(c) Identify Packet Z: _____ *[1]*

(d) Explain what Computer A will do if Packet 1 was completely dropped by a faulty router and Computer B received nothing: *[2]*

Q7. ARQ Timeout Timer Evaluation *[5 Marks]*

(a) Explain what would happen if the sender's timeout timer was configured to be **too short**: *[2.5]*

(b) Explain what would happen if the sender's timeout timer was configured to be **too long**: *[2.5]*

SECTION D: ENCRYPTION, KEYS & DIGITAL CERTIFICATES (10 MARKS)

Q8. Symmetric vs Asymmetric Encryption *[5 Marks]*

(a) Explain what is meant by the **Key Distribution Problem** in symmetric encryption: *[2]*

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(b) Complete the comparison table below: *[3]*

Feature	Symmetric Encryption	Asymmetric Encryption
Number of keys required per user pair		
Key used to decrypt message		
Relative processing speed (Fast / Slow)		

Q9. Asymmetric Key Exchange & Digital Certificates *[5 Marks]*

A student named David wants to purchase books from an online store securely.

(a) Which key will David's browser use to encrypt his credit card number? *[1]*
Answer: _____

(b) Which key will the online store use to decrypt David's credit card number? *[1]*
Answer: _____

(c) State TWO items contained within the store's SSL/TLS Digital Certificate that confirm its authenticity: *[2]*

Item 1: _____

Item 2: _____

(d) State the role of the Certificate Authority (CA): *[1]*

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